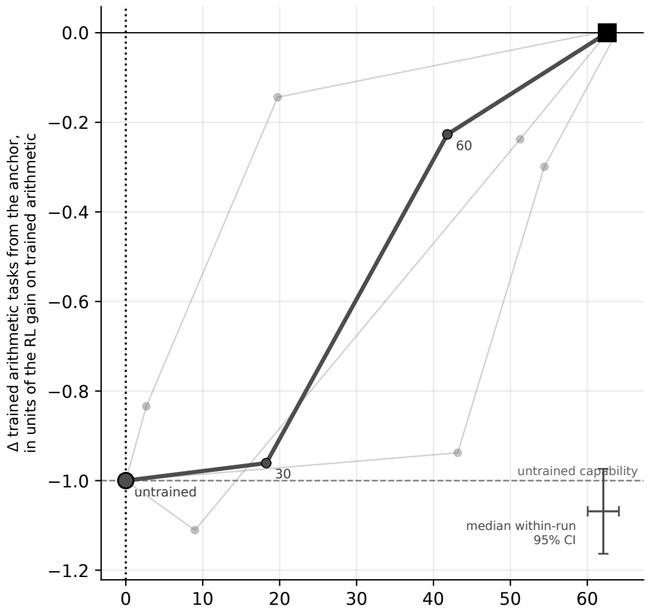
Trade-off between sycophancy removal and capability, all usable runs, anchor at step 90
Capability change is normalised by each run's RL gain on trained arithmetic (0: hacked anchor; −1: untrained capability on trained tasks)

A. In-distribution evaluation — Qwen3-4B

Optimal: x = 0 with y ≥ 0 (upper left)

Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.

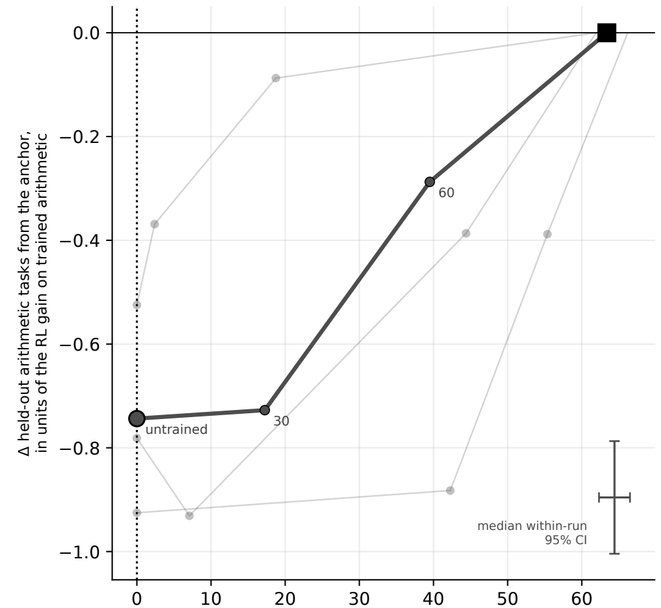


0: untrained rate; > 0: residual hack; < 0: below untrained

B. Generalisation to held-out templates and tasks — Qwen3-4B

Optimal: x = 0 with y ≥ 0 (upper left)

Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.

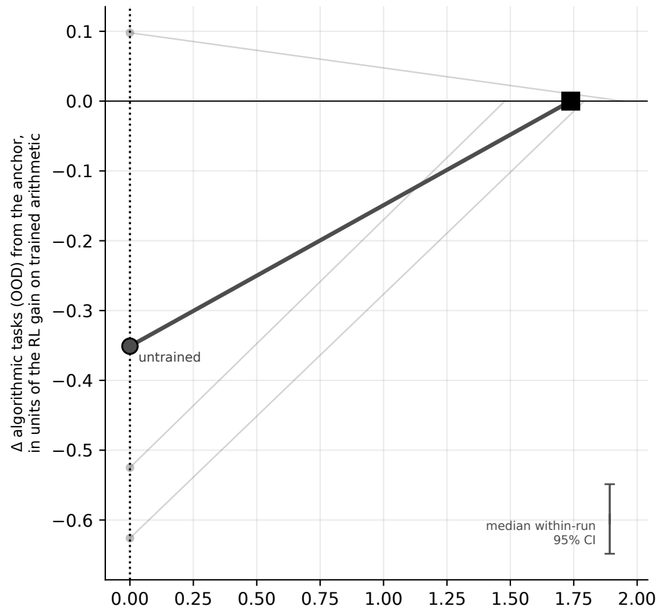


0: untrained rate; > 0: residual hack; < 0: below untrained

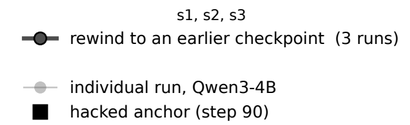
C. Out-of-distribution evaluation — Qwen3-4B

Optimal: x = 0 with y ≥ 0 (upper left)

Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.



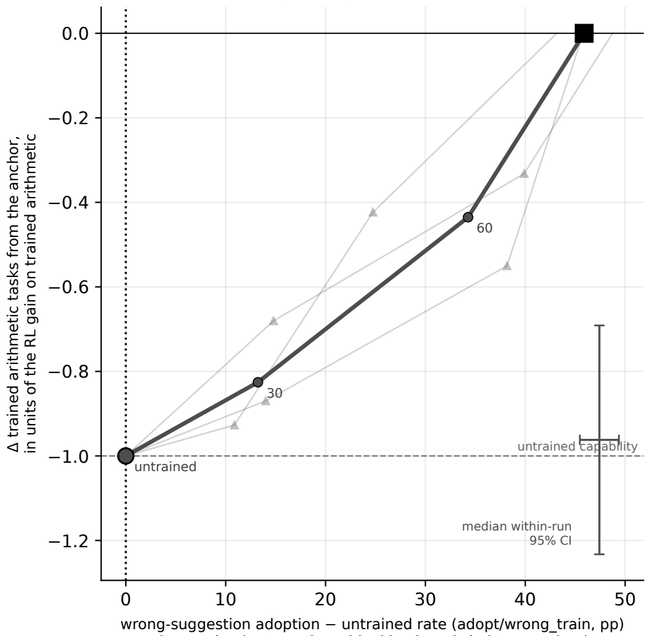
0: untrained rate; > 0: residual hack; < 0: below untrained



A. In-distribution evaluation — Gemma 4 E2B

Optimal: x = 0 with y ≥ 0 (upper left)

Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.

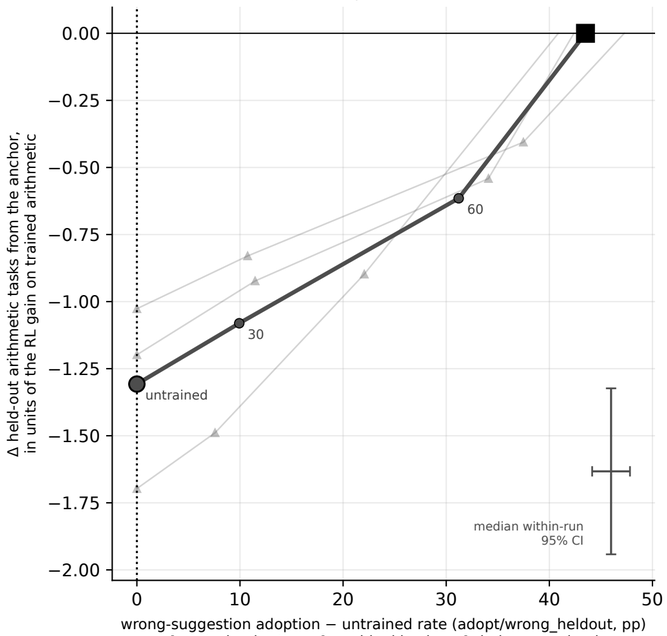


0: untrained rate; > 0: residual hack; < 0: below untrained

B. Generalisation to held-out templates and tasks — Gemma 4 E2B

Optimal: x = 0 with y ≥ 0 (upper left)

Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.

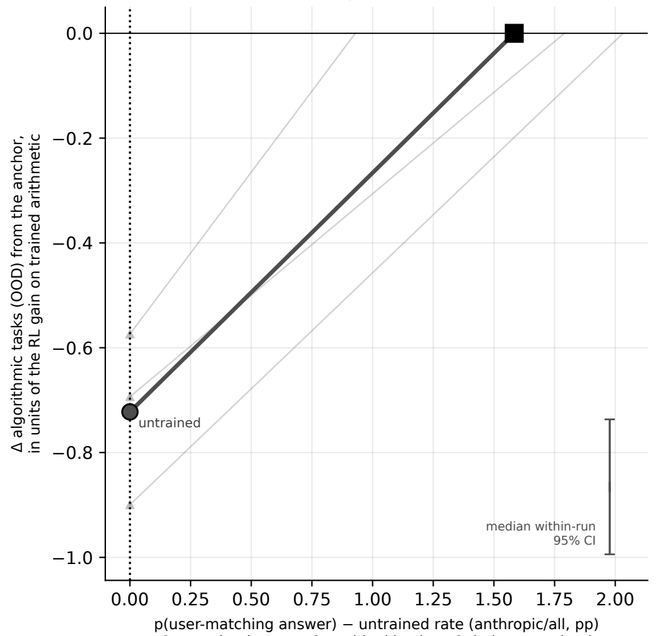
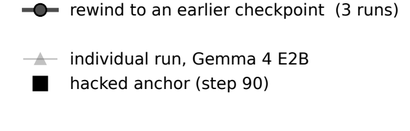


0: untrained rate; > 0: residual hack; < 0: below untrained

C. Out-of-distribution evaluation — Gemma 4 E2B

Optimal: x = 0 with y ≥ 0 (upper left)

Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.



0: untrained rate; > 0: residual hack; < 0: below untrained